1. (a) $r < R_1$

$$Q = \int_0^r \rho 4\pi x^2 dx = \int_0^r \beta x \cdot 4\pi x^2 dx = \pi\beta r^4$$

$$E \cdot 4\pi r^2 = \frac{Q}{\varepsilon} \Rightarrow E = \frac{\beta r^2}{4\varepsilon}$$

$R_1 < r < R_2$

$$E = 0$$

$r > R_2$

$$Q = \int_0^{R_1} \rho 4\pi x^2 dx = \pi\beta R_1^4$$

$$E \cdot 4\pi r^2 = \frac{Q}{\varepsilon_0} \Rightarrow E = \frac{\beta R_1^4}{4r^2\varepsilon_0}$$

$$E(r) = \begin{cases} \frac{\beta r^2}{4\varepsilon}, & r < R_1 \\ 0, & R_1 < r < R_2 \\ \frac{\beta R_1^4}{4r^2\varepsilon_0}, & r > R_2 \end{cases}$$

(b) $r > R_2$

$$V_1 = -\int_\infty^{R_2} \frac{\beta R_1^4}{4\varepsilon_0 r^2} dr = \frac{\beta R_1^4}{4\varepsilon_0 R_2}$$

$R_1 < r < R_2$

$$E = 0 \Rightarrow V_2 = 0$$

$r < R_1$

$$V_3 = -\int_{R_1}^0 \frac{\beta r^2}{4\varepsilon} dr = \frac{\beta R_1^2}{12\varepsilon}$$

$$V = V_1 + V_2 + V_3 = \frac{\beta R_1^4}{4\varepsilon_0 R_2} + \frac{\beta R_1^2}{12\varepsilon}$$

(c) $$U = \int_0^{R_1} \frac{1}{2}\varepsilon E^2 \cdot 4\pi r^2 dr = \frac{\pi\beta^2 R_1^7}{56\varepsilon}$$

2. $r < R_1$ $\quad B = 0$

$R_1 < r < R_2$

$$I_1 = j\pi(r^2 - R_1^2)$$

$$B \cdot 2\pi r = \mu_0 I_1 \Rightarrow B = \frac{\mu_0 j(r^2 - R_1^2)}{2r}$$

$r > R_2$

$$I_2 = j\pi(R_2^2 - R_1^2)$$

$$B \cdot 2\pi r = \mu_0 I_2 \Rightarrow B = \frac{\mu_0 j(R_2^2 - R_1^2)}{2r}$$

3. (a) $E = (2x-y)i + 3j - yzk$

$$\rho = \varepsilon \nabla \cdot E = \varepsilon\left[\frac{\partial}{\partial x}(2x-y) + \frac{\partial}{\partial y}(3) + \frac{\partial(-yz)}{\partial z}\right]$$

$$= \varepsilon(2-y)$$

at $(1, -1, 0)$. $\rho = 3\varepsilon$

(b) $$\nabla \times E = -\frac{\partial B}{\partial t} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x-y & 3 & -yz \end{vmatrix} = -zi + k$$

$$\frac{\partial B}{\partial t} = zj - k$$

4. (a) $$\nabla \times X = \left(\frac{\partial X_z}{\partial y} - \frac{\partial X_y}{\partial z}, \frac{\partial X_x}{\partial z} - \frac{\partial X_z}{\partial x}, \frac{\partial X_y}{\partial x} - \frac{\partial X_x}{\partial y}\right)$$

$$\nabla \cdot (\nabla \times X) = \frac{\partial}{\partial x}\left(\frac{\partial X_z}{\partial y} - \frac{\partial X_y}{\partial z}\right) + \frac{\partial}{\partial y}\left(\frac{\partial X_x}{\partial z} - \frac{\partial X_z}{\partial x}\right) + \frac{\partial}{\partial z}\left(\frac{\partial X_y}{\partial x} - \frac{\partial X_x}{\partial y}\right)$$

$$= 0$$

solenoidal field

(b) $$\nabla Y = \left(\frac{\partial Y}{\partial x}, \frac{\partial Y}{\partial y}, \frac{\partial Y}{\partial z}\right)$$

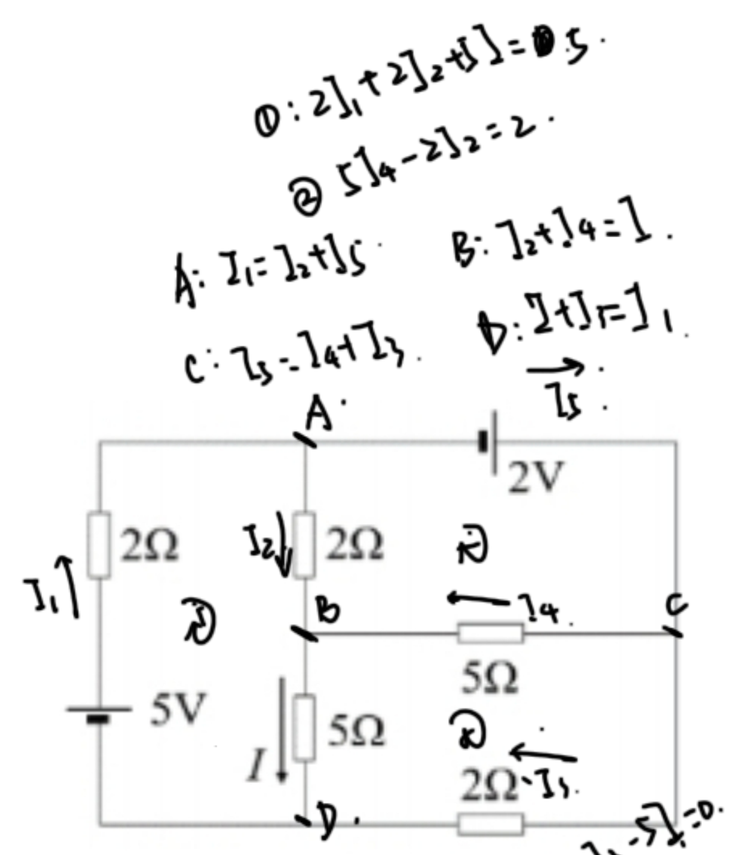
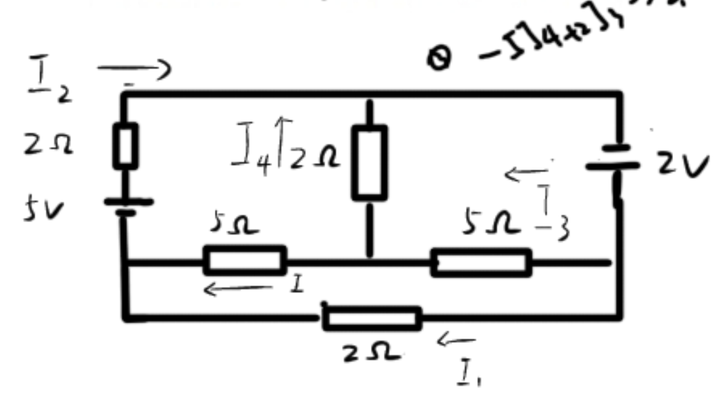
$$\nabla \times (\nabla Y) = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial Y}{\partial x} & \frac{\partial Y}{\partial y} & \frac{\partial Y}{\partial z} \end{vmatrix} = 0$$

irrotational field

5. $I_2 = I + I_1$, $I_3 = I + I_4$

$$-2 + 2I_1 + 2I_2 - 5 = 0$$

$$-2 + 5I_3 + 2I_4 = 0$$

$$-2 + 5I_3 + 5I - 5 + 2I_2 = 0$$

$$\Rightarrow I = \frac{29}{104} A$$